OCP ACCESS TRACKING DASHBOARD

OpenShift Access Tracking & Reporting

Who can reach what, on every cluster: for the people who hold access, and for the quarterly review that must prove it.

Application 5.8.0 · chart 0.70.12 · every screenshot is from the lab on 2026-10-08: the running dashboard, or a report it generated

AGENDA

Two parts

Part 1 · Why: leadership and audit

3–4	The problem, and proof
5–7	Two needs, and the tool
8–11	Self-service and the audit views
12–16	Quarterly evidence

Part 2 · How: the platform team

18	Who sees what
19–22	Architecture
23–24	Operations and limits
25–26	Rolling it out

THE PROBLEM

Access on OpenShift fails silently

9

RoleBindings granting **admin**, **view** and **edit** to groups that have never existed, found on the reference cluster.

oc get rolebinding reports every one of them as healthy.

A binding names a group that does not exist

It looks healthy and grants nobody.

A binding names a person, not a group

Removing them from the directory revokes nothing: it survives offboarding.

A group silently stopped being refreshed

The sync still reports success.

Source: the dashboard's README, the failures it surfaces and the reference cluster's finding

ON THE LAB

The three silent failures, caught

A binding names a group that does not exist

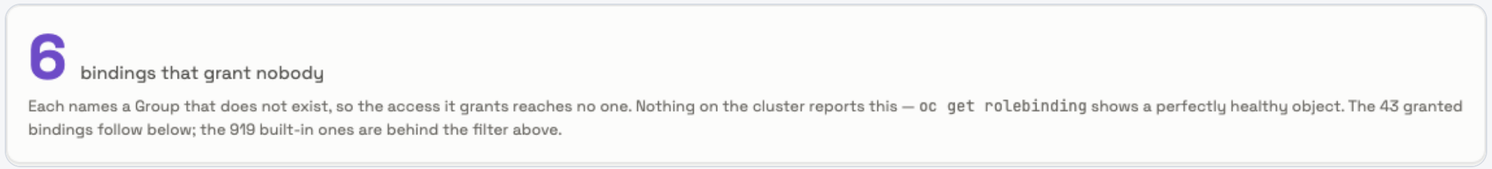
6 bindings that grant nobody, on the host cluster.

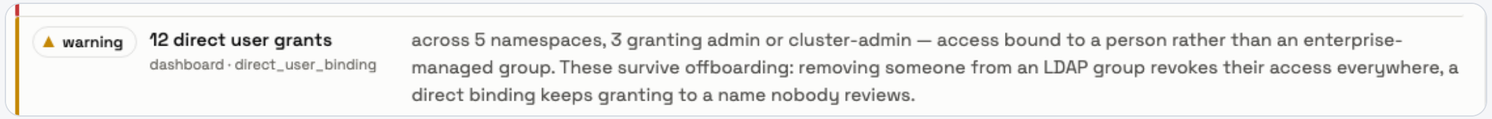
A binding names a person, not a group

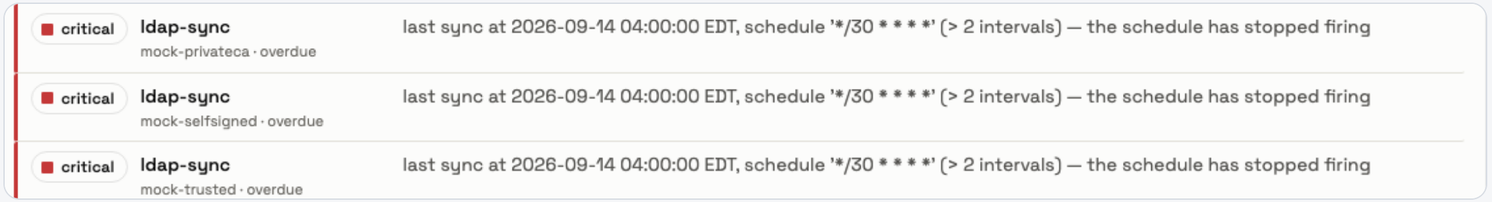
12 direct user grants across 5 namespaces, counting the cluster scope as one; 3 of them admin or cluster-admin.

A group silently stopped being refreshed

ldap-sync overdue on the lab's 3 mock clusters, whose fixture's last sync is fixed at 2026-09-14: how a stopped schedule shows.







Screenshots: Access granted and Overview, light theme, the lab at 5.8.0 on 2026-10-08

WHY NOW

Two needs, one source of truth

Self-service

Everyone can answer “what access do I have, and how did I get it” without a ticket, an oc login with rights they do not have, or a platform engineer.

Quarterly access review

Auditors need evidence that names the access, the reviewer, the time and the outcome, and shows what changed since the last review. SOC 2, ISO 27001 A.5.18 (which replaced A.9.2.5 of the 2013 edition) and SOX centre on dormant, orphaned and privileged access.

The same observed data answers both: one view of access per person, and evidence for the review, drawn from it.

Sources: ACCESS_CONTROL.md (the self tier's purpose) · DESIGN_reporting_service.md (compliance guidance)

THE TOOL

What it is

Read-only

It observes; it never creates or edits a GroupSync CR, a binding or a group.

Every cluster, one view

One dashboard on the host cluster reads every joined cluster, each with that cluster's own token.

Signed in with OpenShift

People log in the way they already do; the cluster decides what each person may see.

Evidence on demand and on schedule

Eleven reports as PDF/A, HTML, CSV and JSON, generated by hand or by a schedule.

Source: README · the chart's templates · the reports catalogue

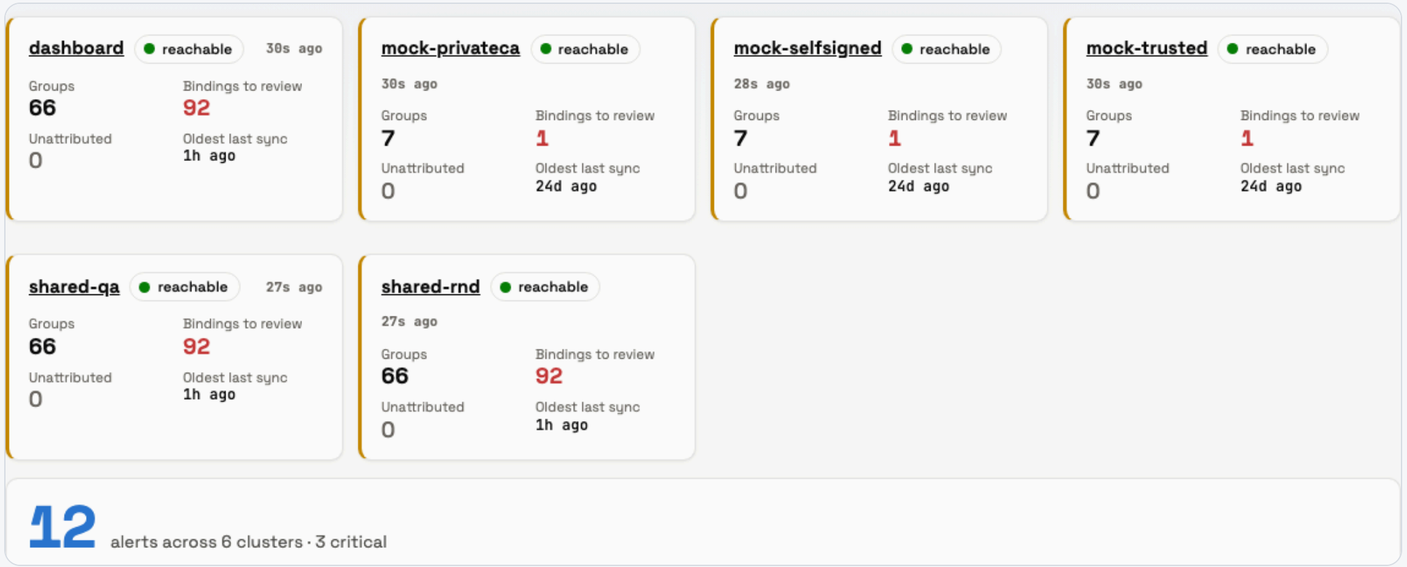
OVERVIEW

Every cluster on one page

One card per joined cluster: reachable or not, its groups, the bindings to review and its oldest sync.

Alerts from every cluster in one list, worst first: 12 across 6 clusters, 3 critical, on the lab.

Each alert says why it fired, and some say what to check: “synced with zero members — check the LDAP-side member DNs resolve”.

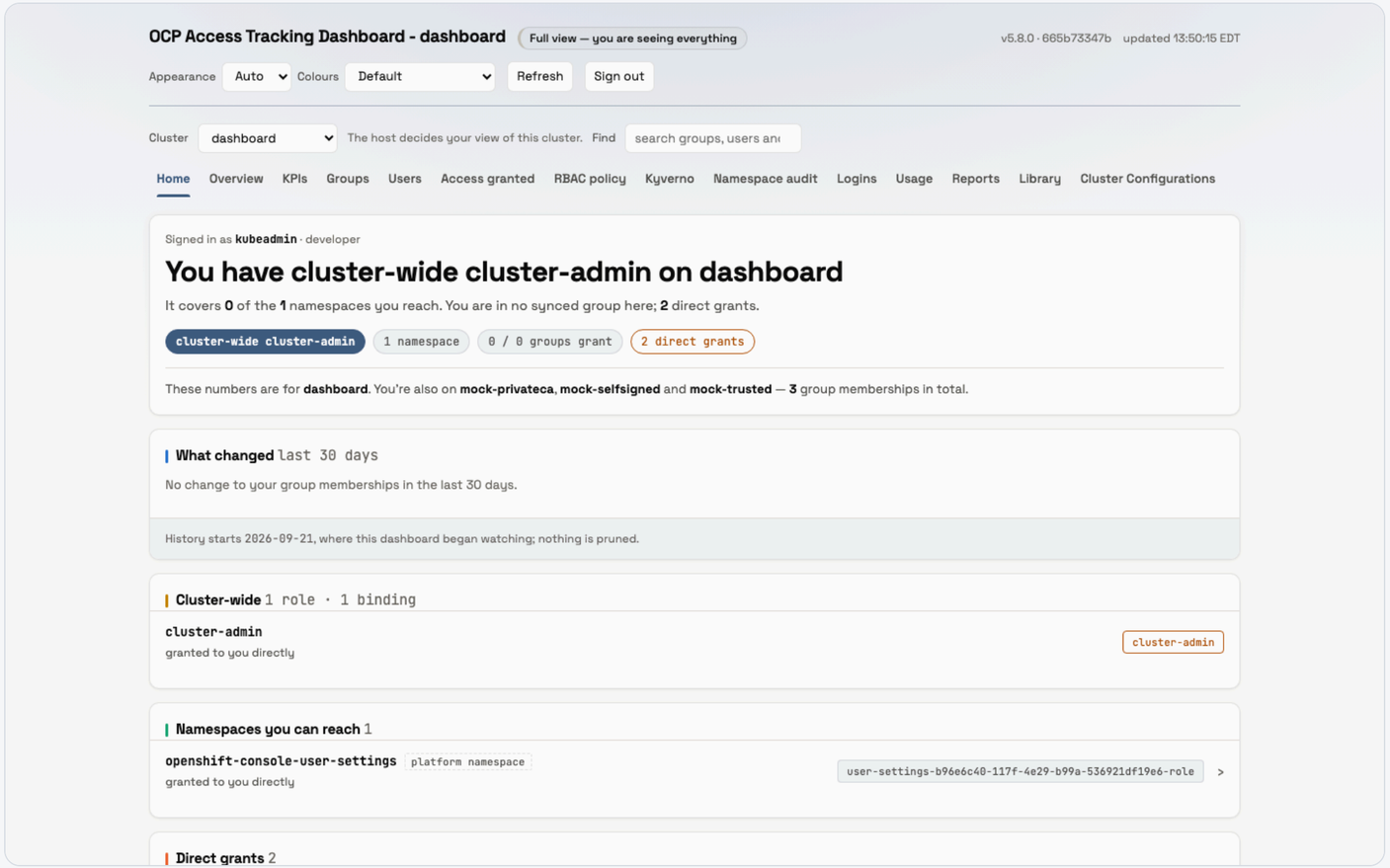


Screenshot: Overview, light theme, the lab at 5.8.0, whose six clusters include three mock API servers

FOR EVERYONE

Self-service: everyone sees their own access

- **Home** is always the reader's own access, administrators included.
- It shows the cluster-wide roles, the namespaces they reach and how: through a group, or a grant naming them directly.
- A view someone may not open shows a named refusal, never a blank page: an empty audit tab would read as a healthy cluster.



Screenshot: Home, light theme, the lab at 5.8.0 · ACCESS_CONTROL.md and the README for the refusal rule

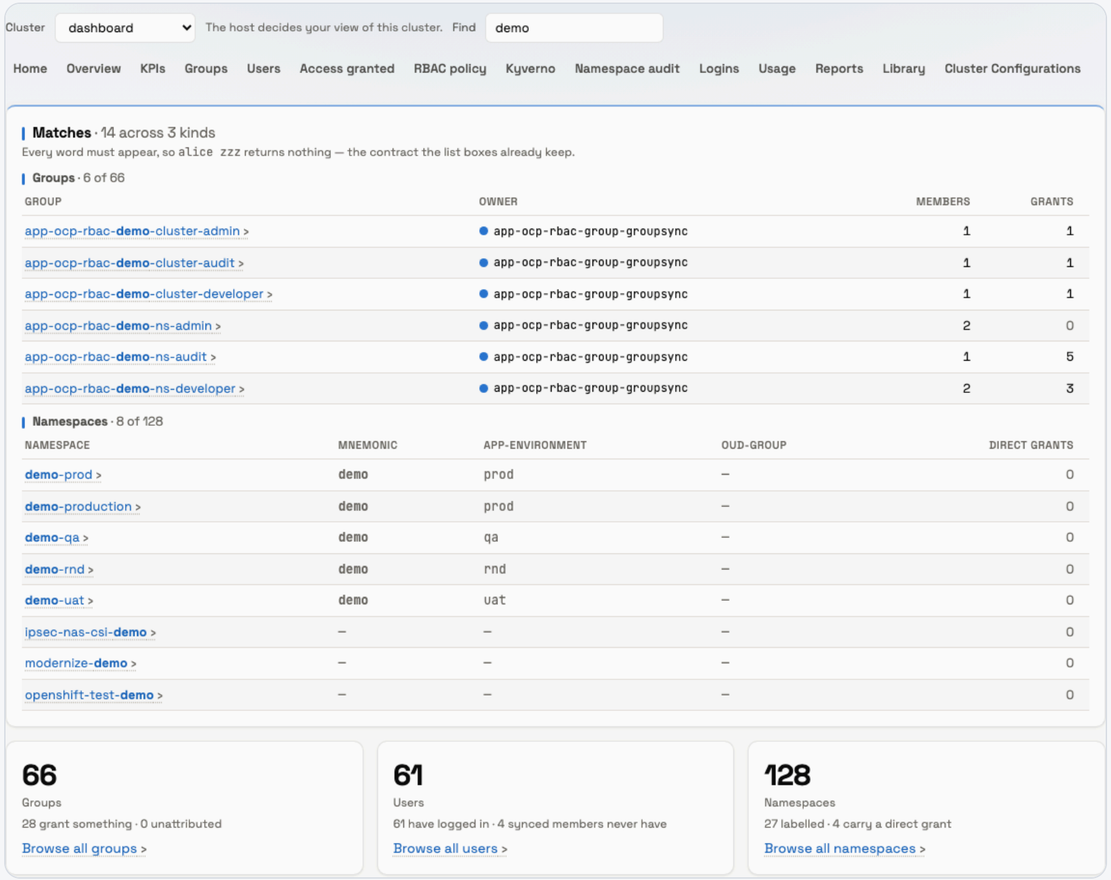
SEARCH

Find anything from one box

One Find box searches groups, users and namespaces together; every word must appear.

“demo” finds 14 matches: 6 groups and 8 namespaces, each a link to its page.

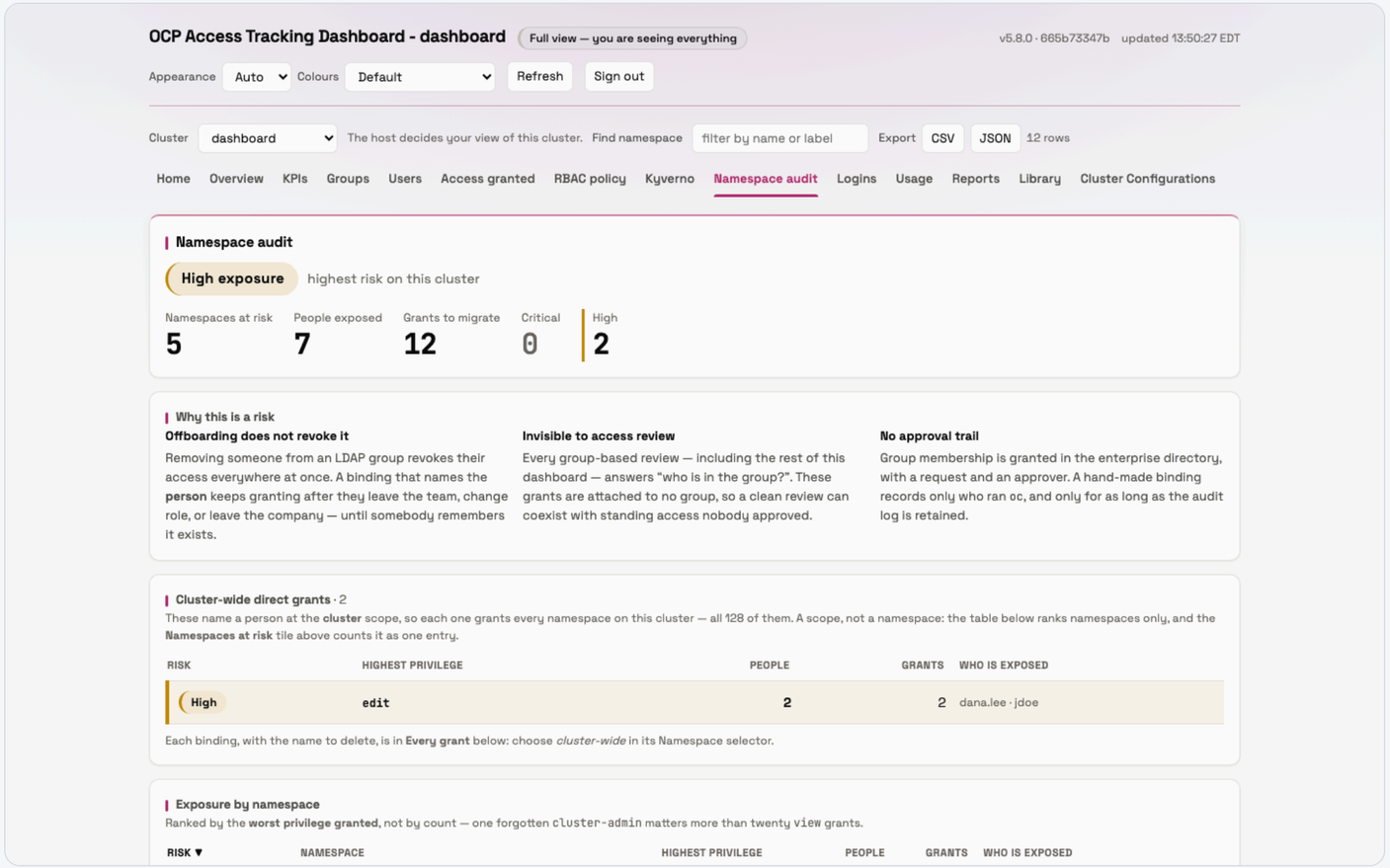
Under the results, the cluster’s totals: 66 groups, 61 users, 128 namespaces, as an auditor sees them. A self-tier reader searches only their own groups, their user row and the namespaces their grants reach.



Screenshot: the Find box, light theme, the lab at 5.8.0

For owners and auditors: what a group review cannot see

- Grants that name a **person** rather than a group survive offboarding, and every group-based review misses them.
- The page ranks namespaces by the worst privilege granted and names who is exposed.
- **Operational goal: zero rows.** At zero, one directory change offboards a person from the entire cluster, and an access review that reads clean is clean.

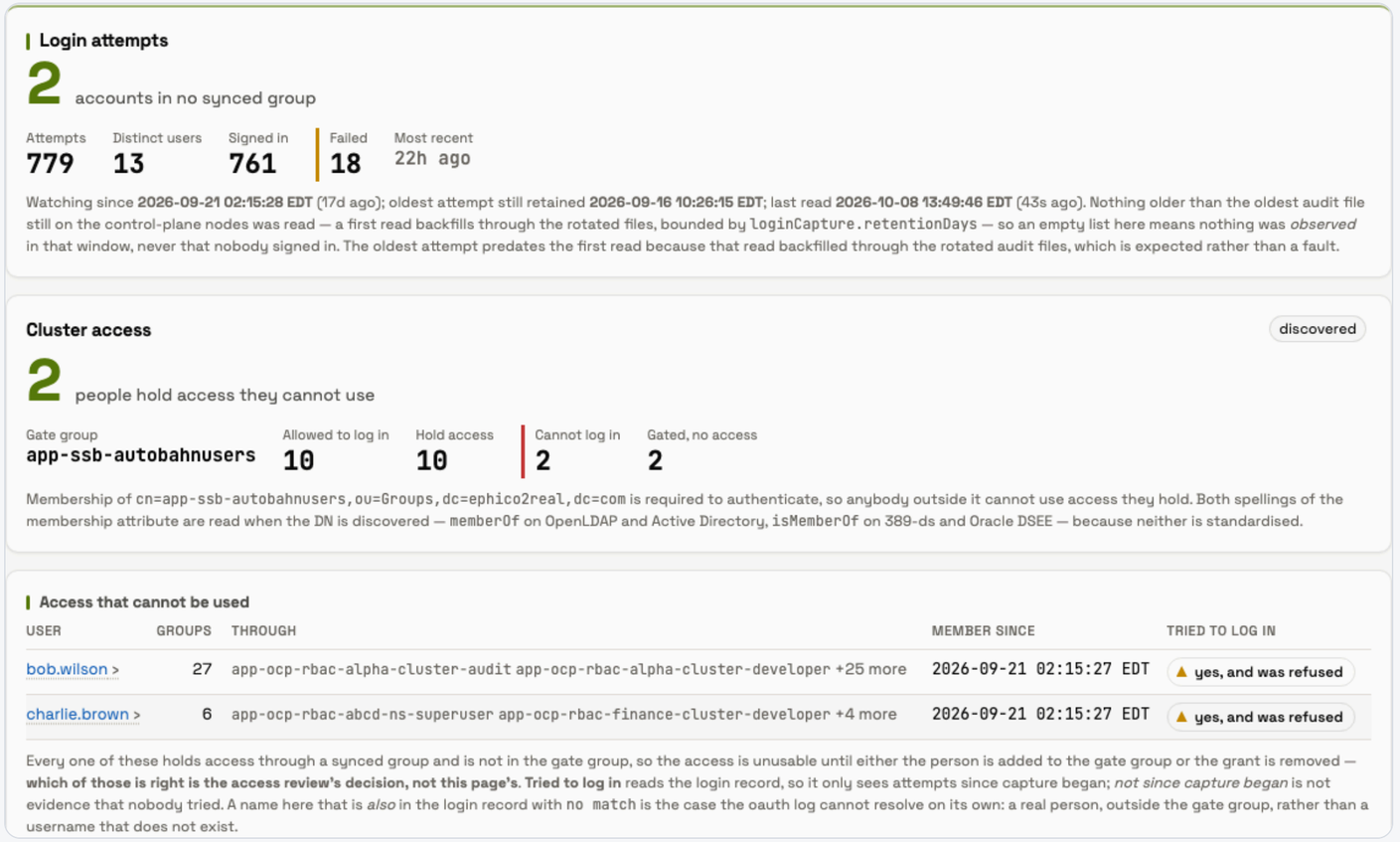


Screenshot: Namespace audit, light theme, the lab at 5.8.0 · the page's own text

LOGINS

Who signs in, and access nobody can use

- Login attempts read from the cluster's OAuth audit log: 13 distinct users made 779 attempts, and 18 failed.
- 2 people hold access through synced groups but are outside the gate group that allows a login, so they cannot use it: a decision for the review.
- Attempts are kept 400 days. Nothing older than the oldest audit file still on the cluster is read, and the page says so.

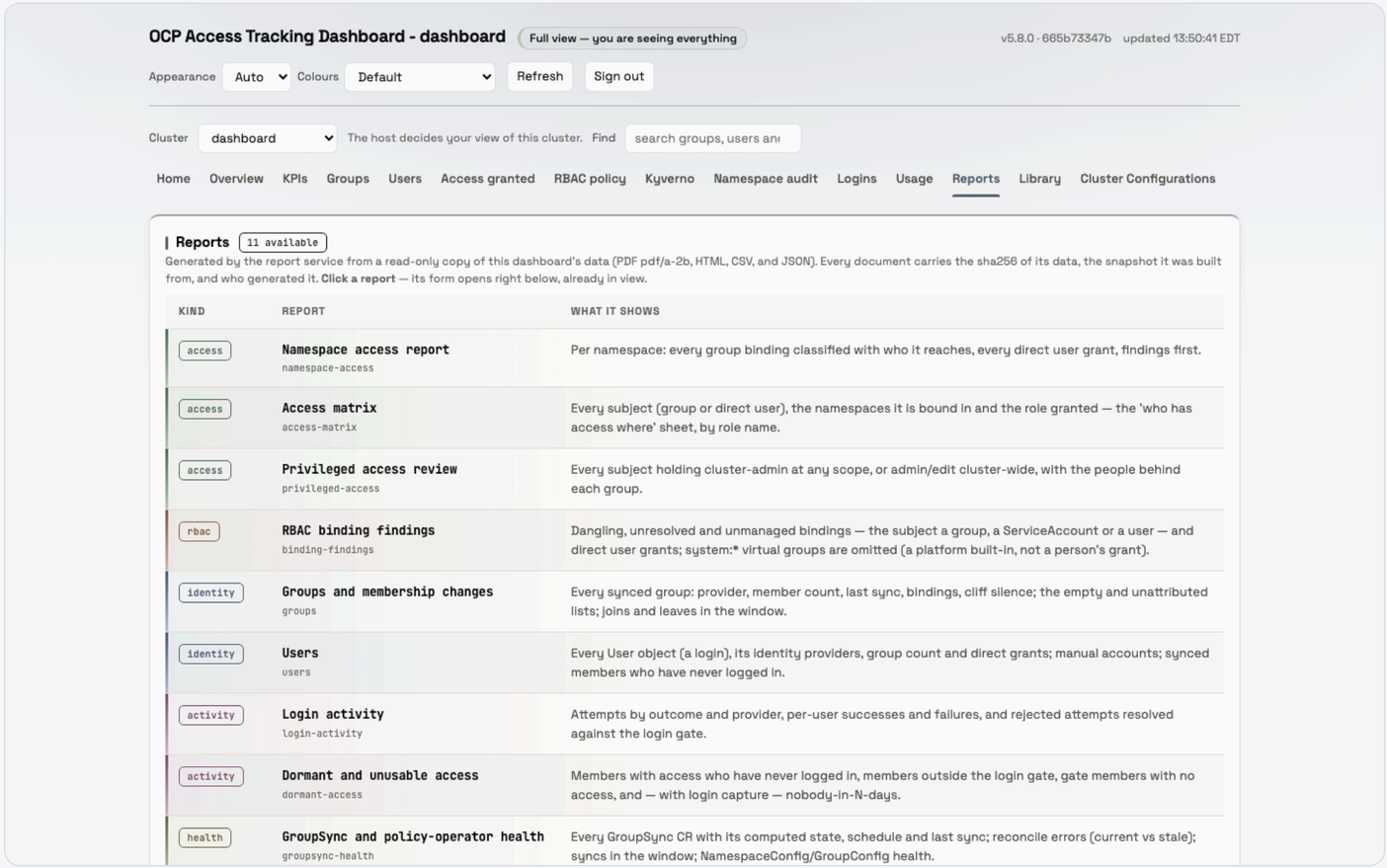


Screenshot: Logins, light theme, the lab at 5.8.0 · values.yaml (login retention)

EVIDENCE

Eleven reports, one per review question

- Who can reach each namespace; the access matrix; privileged access; binding findings.
- Groups and their changes; users; login activity; dormant access; sync health.
- A one-page compliance snapshot, and an access certification pack for sign-off.
- Every document carries the sha256 of its data, the snapshot it was built from, and who generated it.

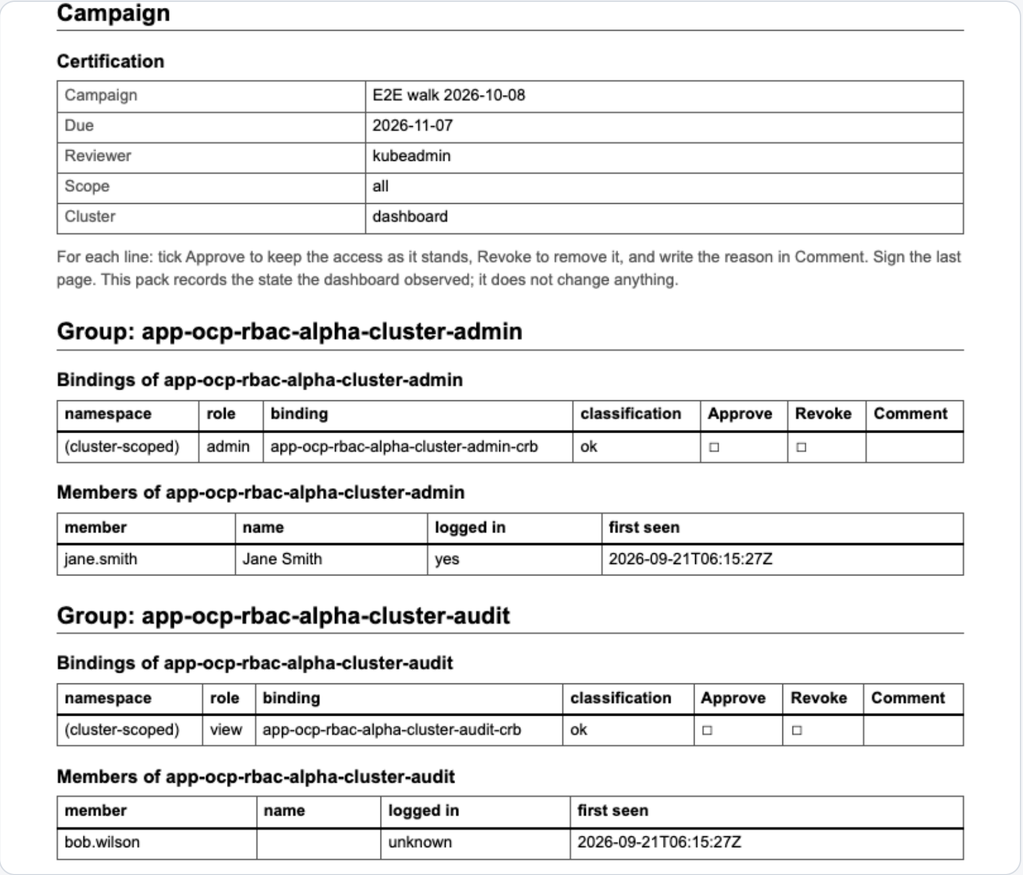


Screenshot: Reports, light theme, the lab at 5.8.0 · docs/reports/README.md

QUARTERLY REVIEW

The certification pack: approve or revoke, line by line

- A campaign with a due date and a named reviewer.
- Every group's bindings, and every grant naming a person directly, with empty **Approve**, **Revoke** and **Comment** columns, and the members behind each group.
- “This pack records the state the dashboard observed; it does not change anything.” The reviewer signs the last page.



Campaign

Certification

Campaign	E2E walk 2026-10-08
Due	2026-11-07
Reviewer	kubeadmin
Scope	all
Cluster	dashboard

For each line: tick Approve to keep the access as it stands, Revoke to remove it, and write the reason in Comment. Sign the last page. This pack records the state the dashboard observed; it does not change anything.

Group: app-ocp-rbac-alpha-cluster-admin

Bindings of app-ocp-rbac-alpha-cluster-admin

namespace	role	binding	classification	Approve	Revoke	Comment
(cluster-scoped)	admin	app-ocp-rbac-alpha-cluster-admin-crb	ok	☐	☐	

Members of app-ocp-rbac-alpha-cluster-admin

member	name	logged in	first seen
jane.smith	Jane Smith	yes	2026-09-21T06:15:27Z

Group: app-ocp-rbac-alpha-cluster-audit

Bindings of app-ocp-rbac-alpha-cluster-audit

namespace	role	binding	classification	Approve	Revoke	Comment
(cluster-scoped)	view	app-ocp-rbac-alpha-cluster-audit-crb	ok	☐	☐	

Members of app-ocp-rbac-alpha-cluster-audit

member	name	logged in	first seen
bob.wilson		unknown	2026-09-21T06:15:27Z

Screenshot: the access-certification report as HTML, generated on the lab at 5.8.0

QUARTERLY REVIEW

A review, start to finish

1

The pack is made

from Reports at the quarter’s start, with the campaign, the reviewer and the due date.

2

The reviewer decides

Approve or Revoke on each line, and signs the last page.

3

The owner revokes

The mnemonic owner removes the person from the group in the central LDAP.

4

GroupSync syncs

On its next run the operator rewrites the OpenShift group from LDAP, which removes them.

5

The dashboard shows it

on its next poll; a report diff of the next pack against this one shows the change.

A grant naming a person directly is not in LDAP, so step 3 cannot remove it.

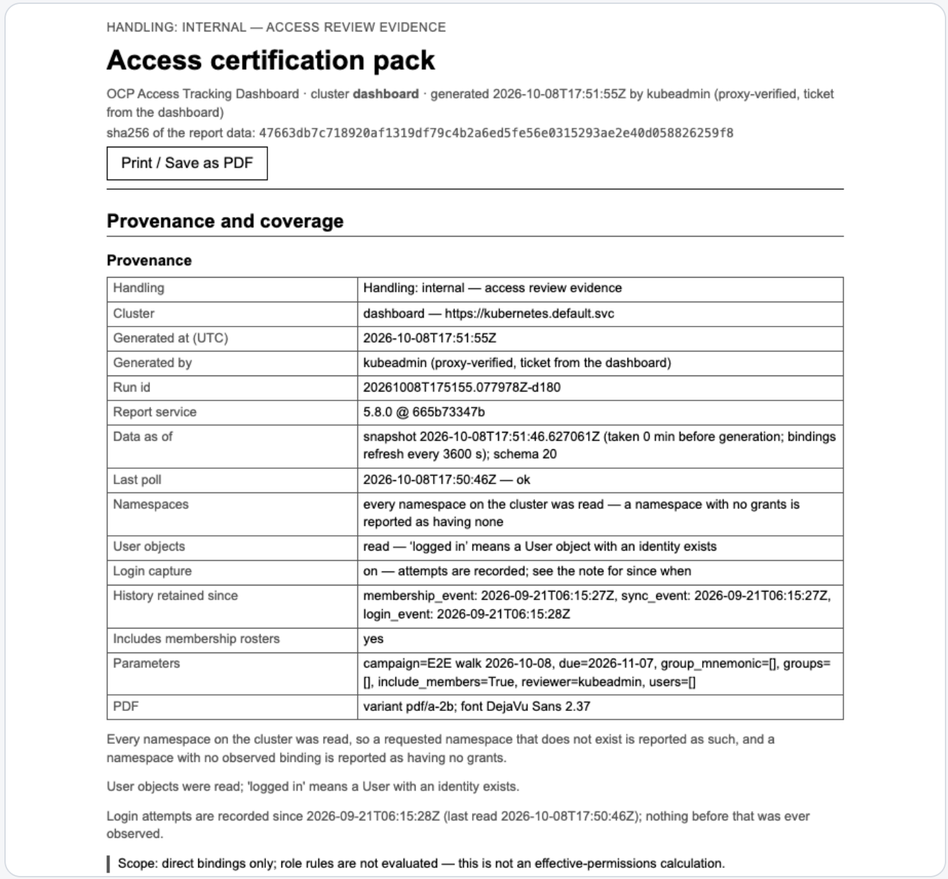
Revoked: delete the binding on the cluster. Approved: add the person to the right LDAP group, wait one sync, then delete the binding. The Namespace audit lists every one.

Sources: reference-architecture.md §1a · the certification report · the lab's GroupSync CRs (every 30 min to 2 h) · SPEC_F3

PROVENANCE

Evidence an auditor can trust

- The sha256 of the report's data, the snapshot it was read from, and its schema.
- Who generated it (a person verified by the OpenShift login, or the schedule by name) and when.
- PDF/A-2b for archiving; a report diff answers “what changed since the last review”.
- Login attempts are kept 400 days, so a year-over-year review still sees the previous cycle.



HANDLING: INTERNAL — ACCESS REVIEW EVIDENCE

Access certification pack

OCP Access Tracking Dashboard · cluster **dashboard** · generated 2026-10-08T17:51:55Z by kubeadmin (proxy-verified, ticket from the dashboard)

sha256 of the report data: `47663db7c718920af1319df79c4b2a6ed5fe56e0315293ae2e40d058826259f8`

Print / Save as PDF

Provenance and coverage

Provenance

Handling	Handling: internal — access review evidence
Cluster	dashboard — https://kubernetes.default.svc
Generated at (UTC)	2026-10-08T17:51:55Z
Generated by	kubeadmin (proxy-verified, ticket from the dashboard)
Run id	20261008T175155.077978Z-d180
Report service	5.8.0 @ 665b73347b
Data as of	snapshot 2026-10-08T17:51:46.627061Z (taken 0 min before generation; bindings refresh every 3600 s); schema 20
Last poll	2026-10-08T17:50:46Z — ok
Namespaces	every namespace on the cluster was read — a namespace with no grants is reported as having none
User objects	read — 'logged in' means a User object with an identity exists
Login capture	on — attempts are recorded; see the note for since when
History retained since	membership_event: 2026-09-21T06:15:27Z, sync_event: 2026-09-21T06:15:27Z, login_event: 2026-09-21T06:15:28Z
Includes membership rosters	yes
Parameters	campaign=E2E walk 2026-10-08, due=2026-11-07, group_mnemonic=[], groups=[], include_members=True, reviewer=kubeadmin, users=[]
PDF	variant pdf/a-2b; font DejaVu Sans 2.37

Every namespace on the cluster was read, so a requested namespace that does not exist is reported as such, and a namespace with no observed binding is reported as having no grants.

User objects were read; 'logged in' means a User with an identity exists.

Login attempts are recorded since 2026-09-21T06:15:28Z (last read 2026-10-08T17:50:46Z); nothing before that was ever observed.

Scope: direct bindings only; role rules are not evaluated — this is not an effective-permissions calculation.

Screenshot: the report's provenance, the lab at 5.8.0 · values.yaml (login retention) · SPEC_F3 (report diff)

EVERY QUARTER

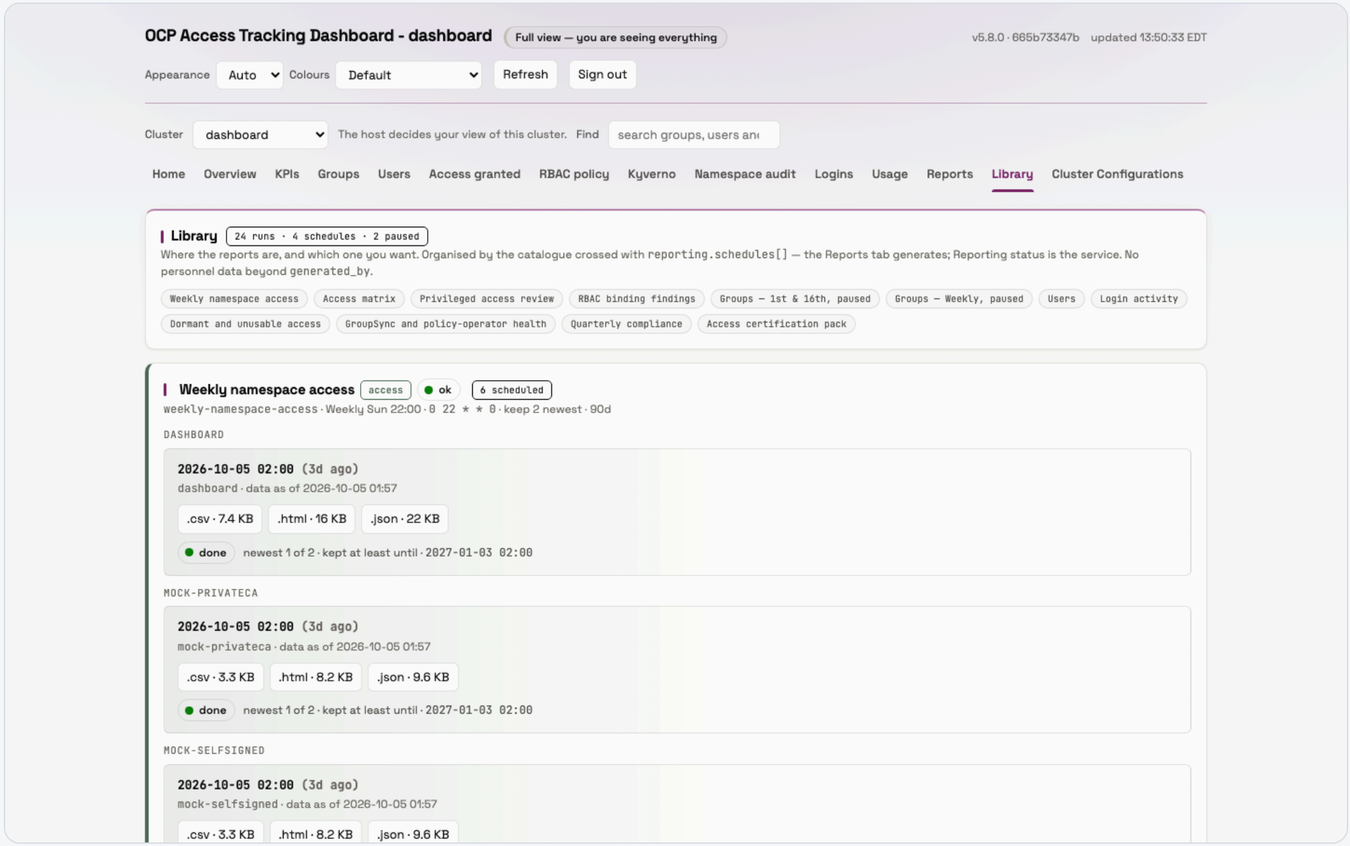
Scheduled, not remembered

quarterly-compliance

```
0 5 1 1,4,7,10 *
```

05:00 New York time on 1 January, April, July and October; every run kept 2,555 days, about seven years, and the newest 12 per cluster beyond that.

- Each scheduled run can send one CloudEvent to a webhook.
- An alert fires when a schedule runs late.



Screenshot: Library, light theme, the lab at 5.8.0 · environments/crc.yaml (the quarterly schedule)

PART 2

How it works

Read-only by design: it reads clusters, and it never changes anyone's access.

ACCESS TIERS

Who sees what: the cluster decides

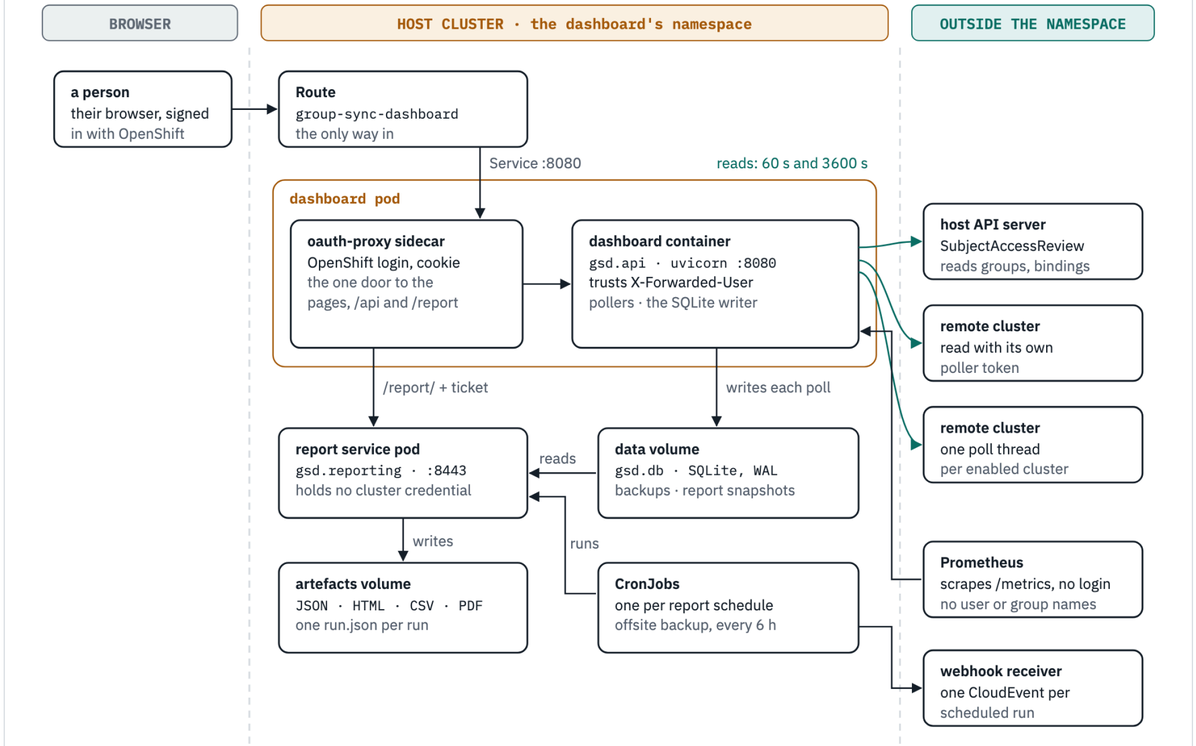
Tier	Who has it	What they see
Self	Everyone who can sign in	Their own access: Home, and their own rows
Auditor	Can list clusterrolebindings, for example through an auditor group	Every cluster-wide audit view, Reports and Library; Usage stays their own
Usage	Passes the usage question; by default the same one as Cluster-admin	The Usage tab: everyone's activity
Cluster-admin	Can update clusterrolebindings	KPIs and Cluster Configurations, plus all of the above

A SubjectAccessReview on the cluster decides each tier; the answer is cached 60 s, and any failure falls back to Self.

Sources: ACCESS_CONTROL.md · values.yaml (adminSar, usageAdminSar, clusterAdminSar) · kube.py TierResolver

ARCHITECTURE

Architecture at a glance



All a person sees enters through the oauth-proxy. Clusters are read; the one default write is the pod's own Lease.

Solid arrows are traffic, curves are one dashboard reading many clusters. Nothing on this page is proposed.

Figure 1 of docs/guides/architecture-overview.md, drawn from the code at 5.8.0

ARCHITECTURE

Inside one poll cycle

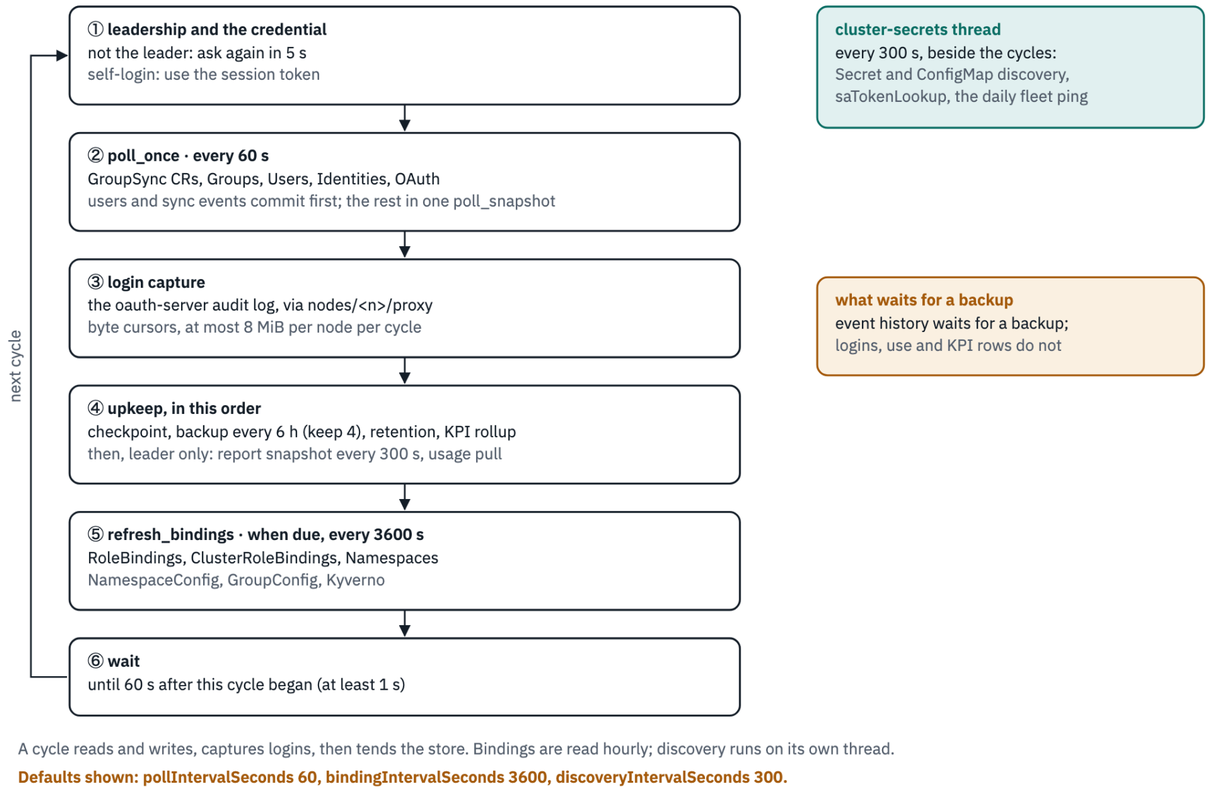


Figure 3 of docs/guides/architecture-overview.md

ARCHITECTURE

How a report is made

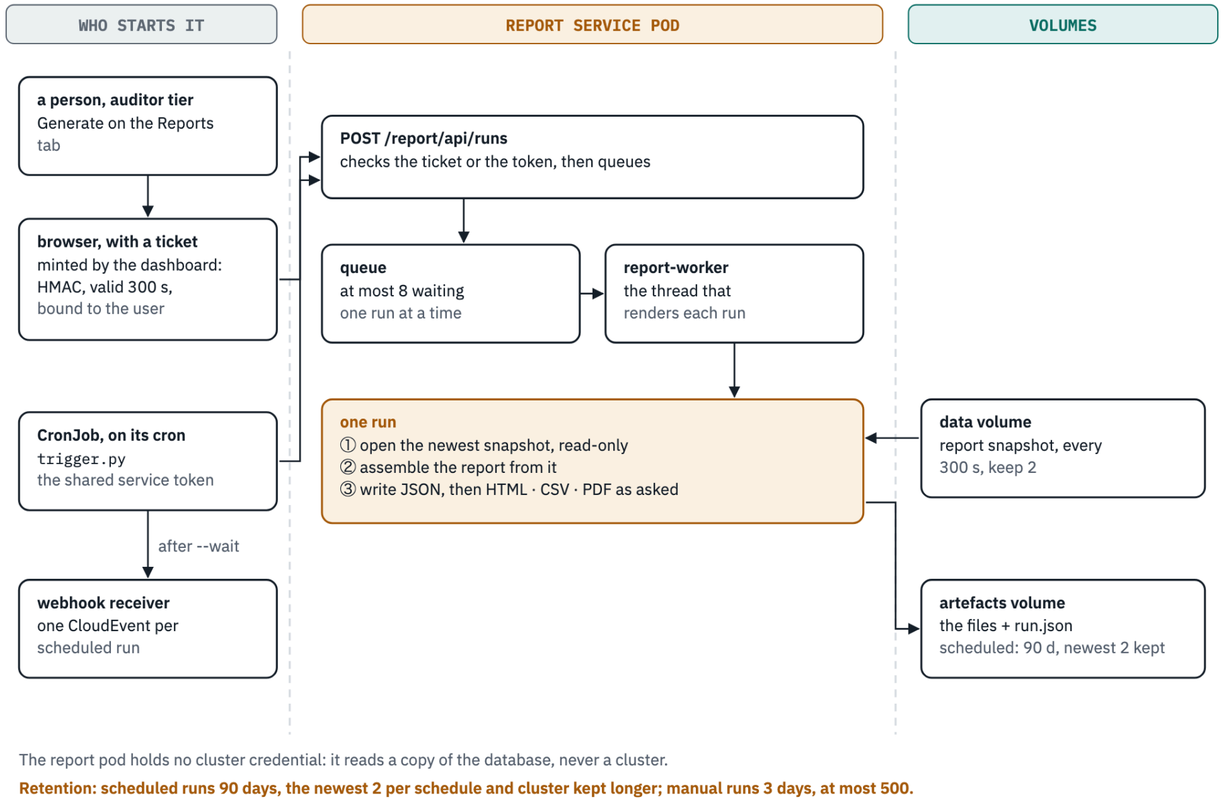
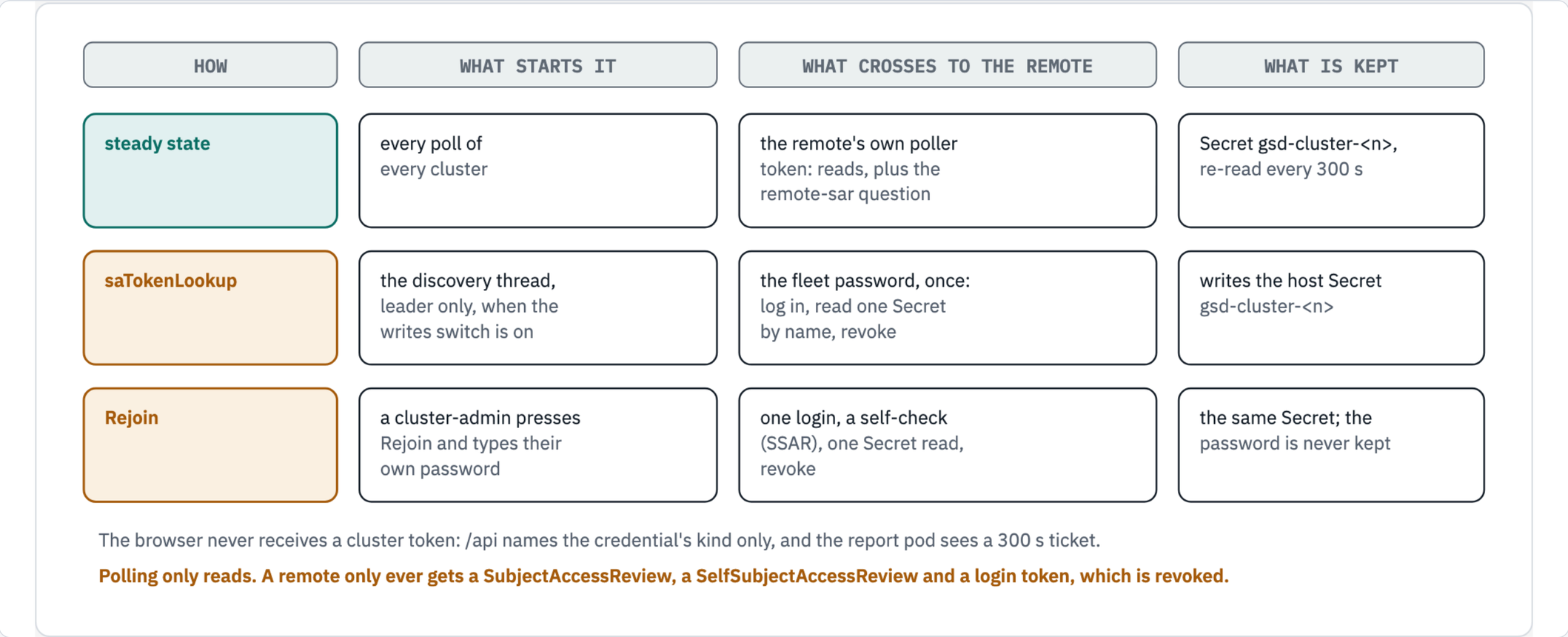


Figure 5 of docs/guides/architecture-overview.md

SECURITY

Credentials and trust boundaries



HOW	WHAT STARTS IT	WHAT CROSSES TO THE REMOTE	WHAT IS KEPT
steady state	every poll of every cluster	the remote's own poller token: reads, plus the remote-sar question	Secret gsd-cluster-<n>, re-read every 300 s
saTokenLookup	the discovery thread, leader only, when the writes switch is on	the fleet password, once: log in, read one Secret by name, revoke	writes the host Secret gsd-cluster-<n>
Rejoin	a cluster-admin presses Rejoin and types their own password	one login, a self-check (SSAR), one Secret read, revoke	the same Secret; the password is never kept

The browser never receives a cluster token: /api names the credential's kind only, and the report pod sees a 300 s ticket.

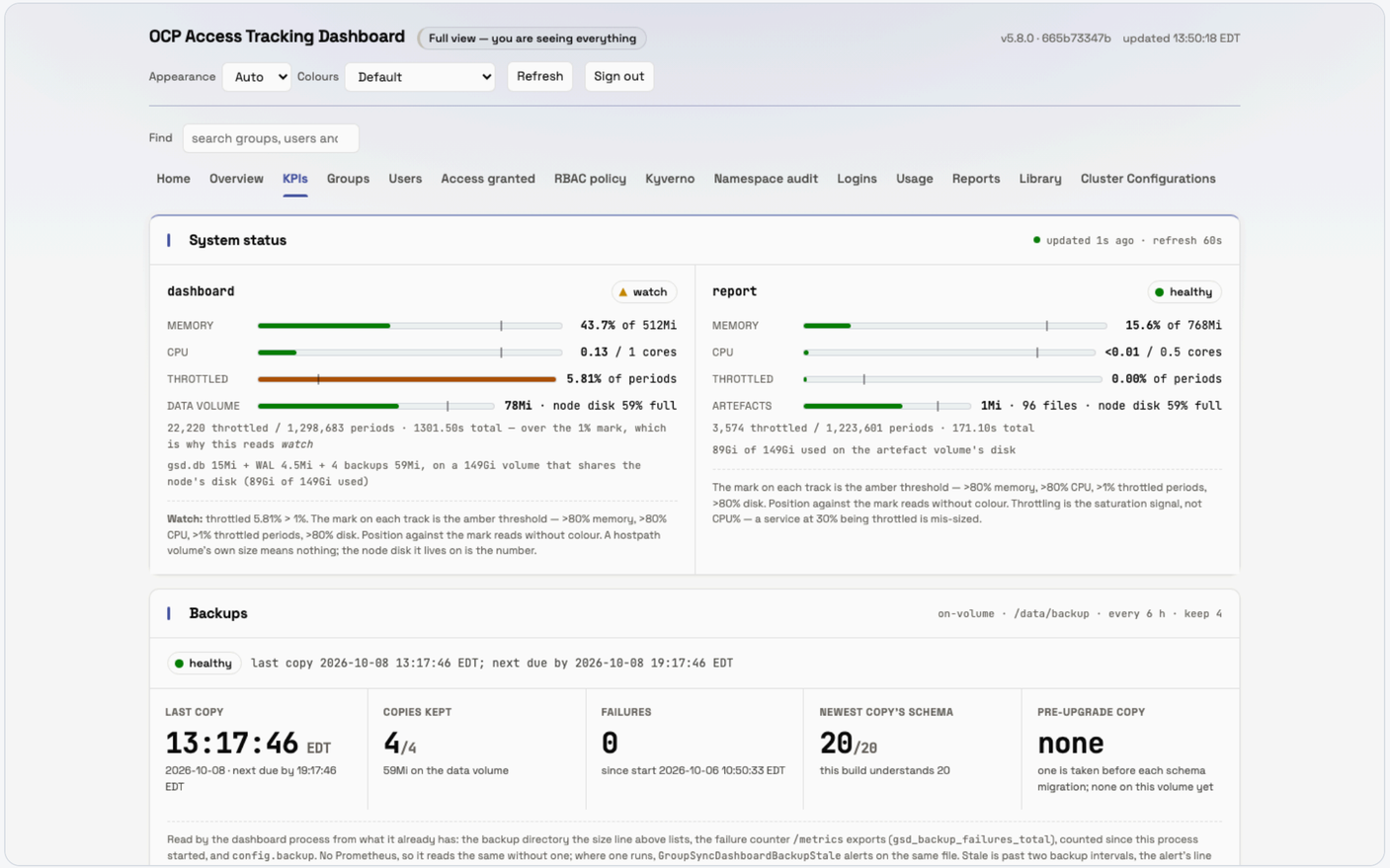
Polling only reads. A remote only ever gets a SubjectAccessReview, a SelfSubjectAccessReview and a login token, which is revoked.

Figure 6 of docs/guides/architecture-overview.md

OPERATE

Operations: what the platform team runs

- One Helm chart, ready for Argo CD; one dashboard pod and one report pod.
- A database backup every 6 h with 4 kept, and an offsite copy every 6 h.
- Twenty Prometheus alerts, from a cluster that stops answering to a backup or a schedule running late.
- The KPIs page shows the system's own health: memory, CPU, throttling, disk, backups.



Screenshot: KPIs, light theme, the lab at 5.8.0 · the chart's PrometheusRule and backup settings

LIMITS

What it does not do

Direct bindings only

It reads who was granted which role. It never expands a role's rules, so it is not a list of every permitted action.

It changes no access

It never creates, edits or deletes a GroupSync CR, a group or a binding. Fixes happen in LDAP or on the cluster.

Login history has a start

Logins go back only to the oldest audit file on the cluster when capture began; nothing earlier was observed. They are kept 400 days.

As of the last read

Groups and CRs are read every 60 s and bindings every hour, so a change shows on the next read.

Sources: reference-architecture.md (What it is not; the cycle) · the reports' coverage notes · values.yaml

NEXT STEPS

Rolling it out

1

Install

the chart on the host cluster, with Helm or Argo CD.

2

Join clusters

each with its own token: a Secret, saTokenLookup or Rejoin.

3

Name the auditors

the groups that review access everywhere.

4

Schedule the evidence

a quarterly compliance schedule, and a webhook for delivery.

No access changes on any cluster: the dashboard reads, and the reports record what it read.

Its only writes are its own: a leader Lease on the host and, when a cluster joins by saTokenLookup or Rejoin, that cluster's token Secret on the host and a login on the remote that it revokes.

Sources: the chart's values.yaml · CLUSTER_CREDENTIALS.md

THANK YOU

Questions, and where to read more

`architecture-overview.md`	the architecture in seven figures
`reference-architecture.md`	the full reasoning, claim by claim
`ACCESS_CONTROL.md`	who sees what, and why
`docs/reports/README.md`	the eleven reports and their schedules

All documents are in the group-sync-dashboard repository, under docs/